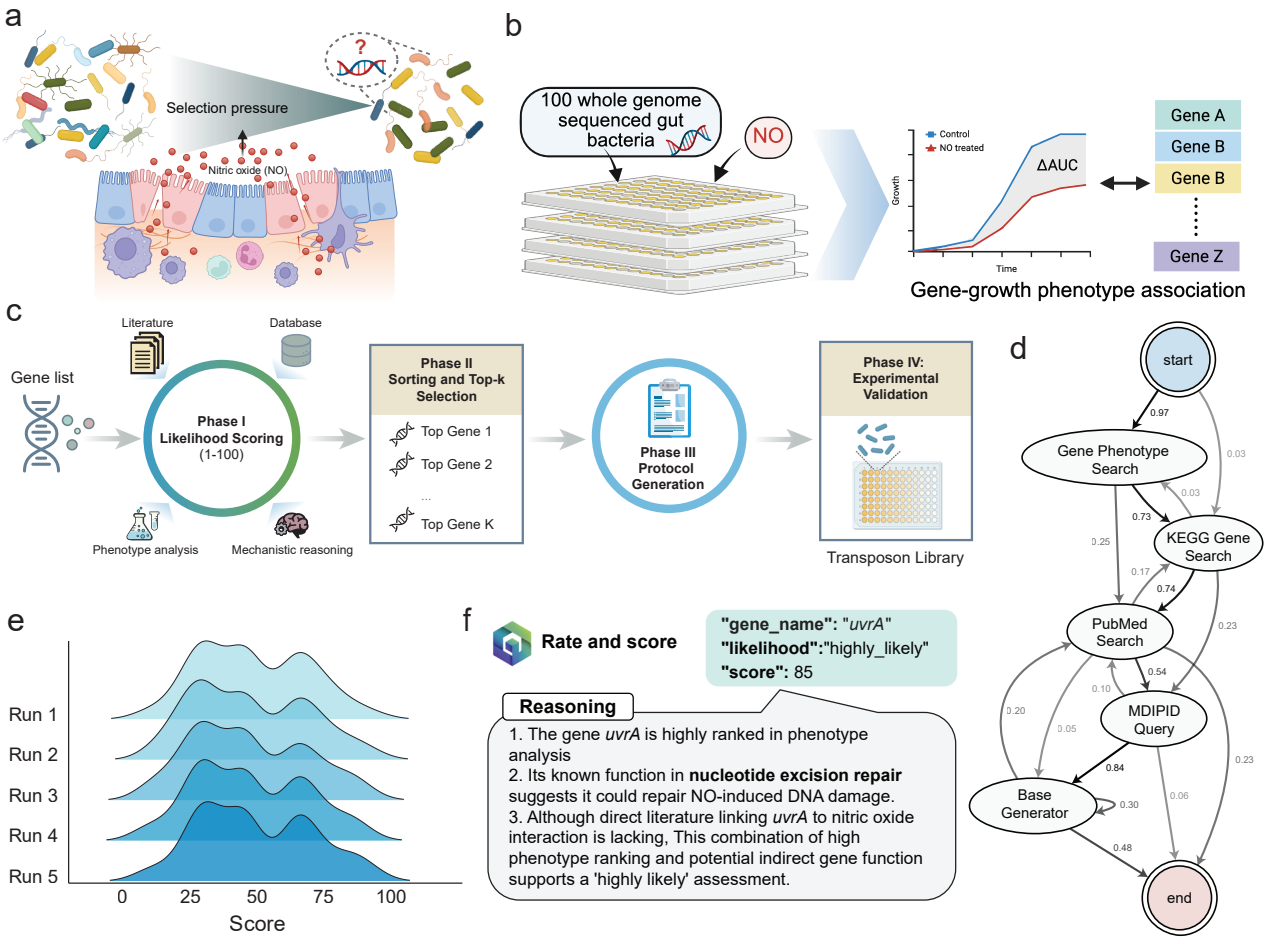
a
Selection pressure
Nitric oxide (NO)
b
100 whole genome sequenced gut bacteria
NO
Control
NO treated
Growth
Time
ΔAUC
Gene A
Gene B
Gene B
Gene Z
Gene-growth phenotype association
c
Literature
Database
Gene list
Phase I
Likelihood Scoring
(1-100)
Phenotype analysis
Mechanistic reasoning
Phase II
Sorting and Top-k Selection
Top Gene 1
Top Gene 2
...
Top Gene K
Phase III
Protocol Generation
Phase IV:
Experimental Validation
Transposon Library
d
start
Gene Phenotype Search
KEGG Gene Search
PubMed Search
MDIPID Query
Base Generator
end
0.97
0.03
0.03
0.73
0.25
0.17
0.74
0.23
0.54
0.10
0.20
0.05
0.84
0.23
0.30
0.06
0.48
e
Run 1
Run 2
Run 3
Run 4
Run 5
0
25
50
75
100
Score
f
Rate and score
"gene_name": "uvrA"
"likelihood":"highly_likely"
"score": 85
Reasoning
1. The gene uvrA is highly ranked in phenotype analysis
2. Its known function in nucleotide excision repair suggests it could repair NO-induced DNA damage.
3. Although direct literature linking uvrA to nitric oxide interaction is lacking, This combination of high phenotype ranking and potential indirect gene function supports a 'highly likely' assessment.